

John V. J. Lang

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AI ARCHITECT | EMBEDDED ENGINEER | GAME & 3D SYSTEMS BUILDER

Hands-on engineer with 8+ years building high-performant ML, distributed data, and AI-native products. I've built multi-million dollar pipelines, cut costs just as much, and grown departments from myself to 20; I now want to follow my passions: games and robots. I rapidly turn unstructured data and 2D designs into validated, production-ready systems and game assets, with a record of reducing cost, accelerating compute, and leading delivery from prototype to release.

RECENT EXPERIENCE

AI Manager

Jan 2026 - Present

AIA Contract Documents: Las Vegas, NV

- Built AIA Contract Documents' first conversational AI and legal-risk platform for clients. Built on continuous context understanding, vector retrieval, application, individual company needs, HITL relearning, and user needs. Scaled chat feature for WebGL 3D construction from prompt and template images.
- Combining MCP tools, enterprise data connectors, and company wide Azure resources so internal teams can interrogate contracts and operational data
- Integrated and evaluated foundation models, Hugging Face prompt-injection guardrails, governed training/evaluation data, document retrieval pipelines, and Copilot-assisted architecture review to improve answer quality, safety, processing speed, and delivery throughput. Established an organization-wide AIDLC

Sr. Spark Software Engineer Team Lead → Data Engineering & Science Manager → AI Architect

Feb 2022 - Jan 2026

Valenz Health: Nashville, TN

- Led hands-on AI and data architecture across 42 engineers and analysts; replaced \$600K/year in siloed tooling and converted analyst/data-engineering codebases to a metadata-driven transformation framework. Built 4 new revenue based pipelines in dynamic, scalable, HITL CV, OCR, and deep learning.
- Identified \$3.2M+ in new business lines and \$750K in annual technology savings while delivering predictive products growing 300% year over year.
- Migrated petabytes of healthcare claims to distributed cloud platforms, model readiness through architected ML Flow, and company wide monitoring..

Data Scientist - Life Sciences and Operations | Healthgrades

Jan 2021 - Feb 2022

- Owned five big-data pipelines and four predictive/NLP systems serving 4,000 daily users; cut workflow runtime 93% and cloud spend by \$78K/month.

Quantitative Analyst II – Product Development | Asurion

May 2020 - Nov 2020

- Built ML pricing and anomaly-detection workflows for IoT products. Reducing a two-week reporting cycle to daily and saving \$100K/year.

AI Strategy Analyst – Products | Churchill Downs

Sep 2018 - May 2020

- Led AI, automation, VR build, and product analytics across five venues; built transaction controls that identified \$85K/year in cash-flow loss.

EDUCATION

B.A. Mathematics & Minor Economics: Wabash College

M.S. Data Science: Eastern University

CERTIFICATIONS

Azure Architect Certified (exp) – Databricks Data Engineer Professional (exp) – Snowflake SnowPro Advanced Architect (exp) – Data Science: Statistics & Machine Learning – Strategic Leadership and Management

PUBLICATION & WHITEPAPER

Repay, Grover, Shipman, Lang, & Tarter. **Polysorbate 80: An Emulsifying Agent in Cannabis Concentrate**

Microbiology. Cannabis Science and Technology: 20-26. 2024

Lang, J. **A Data Intelligence Company Built on the Premise to Never Run Anything Twice.** Whitepaper: Valence

TECHNICAL SKILLS

AI/ML: Python, machine learning, LLMs, Hugging Face, Azure AI Foundry, RAG/vector databases, prompt-injection defense, model evaluation, GCP, AWS
Game/3D: Blender, Unreal Engine, IFCOpenshell, procedural geometry, computer vision, mesh/UV/collision/LOD optimization, vector/dwg/raster extraction
Embedded Systems: ROS 2, SciPy, gpiozero, Picamera2, PyCoral, RPi.GPIO, spidev, MicroPython, RaspberriPi, CoralTPU, CUDA
Systems: Rust, OpenCV, scikit-learn, pytorch, Go, R,, Spark, distributed computing, Snowflake, Databricks, cloud architecture, MCP

SELECTED AI & GAME ENGINEERING

Return-of-the-Mesh: Building an AI-assisted 2D-to-3D pipeline using vector geometry extraction/computer vision/ML, Blender, and Unreal to convert architectural plans and product imagery into dimensionally accurate, game-ready FBX assets. Scales for unstructured DWG, vector PDFs, and Images..

Holoprojector: Transforming images, product urls, specifications, or prompt details from 2D to 3D for Unreal. Validation for closed manifold geometry, wall thickness, topology stitching, pivots, 1 m UV density, collision, LOD/Nanite readiness, and runtime performance.

PROFESSIONAL DEVELOPMENT

Technical Advisor – Technical Advisory Board Member for Artie (Y Combinator 2023)

Rocket PSA – Software building simple, accessible, AI technology for new MSP market entrants served by Digital Ocean

Personal Development – AI and Data Consultant – Democratizing AI, technology, and data engineering for individual/company needs philanthropically
– Henderson Hospital Bi-Weekly Volunteer for patient safety